

Activity 1.1.6

Digital Component Identification

Combinational gates, a D flip-flop, and a 555 astable clock — identified by behavior, built in Multisim, and measured.

Activity Scope

Identify digital logic components by their input/output behavior, build each in simulation, and record the relationships that define them.

01

Combinational logic

Six 74LS gates, problems 1–6. Output set entirely by present inputs.

02

Sequential logic

74LS74N D flip-flop, problem 7. Output depends on clock history.

03

Clock source

LM555CN astable, problem 8. Three circuits differing only in C_2 .

Software and Components

PART	COMPONENT
Logic ICs	74LS00N · 02N · 04N · 08N · 32N · 74N · 86N
Timer	LM555CN
Resistors	100 Ω, 330 Ω
Capacitors	10 nF, 6 / 12 / 24 μF
Switches	SPDT, keyed for live toggling
Indicator	PROBE_DIG_RED

5.00 V

DC supply, all circuits

2.50 V

Digital probe threshold

All circuits built and simulated in NI Multisim. SPDT switches drive gate inputs between GND (logic 0) and +5 V (logic 1).

The Multisim Sheet

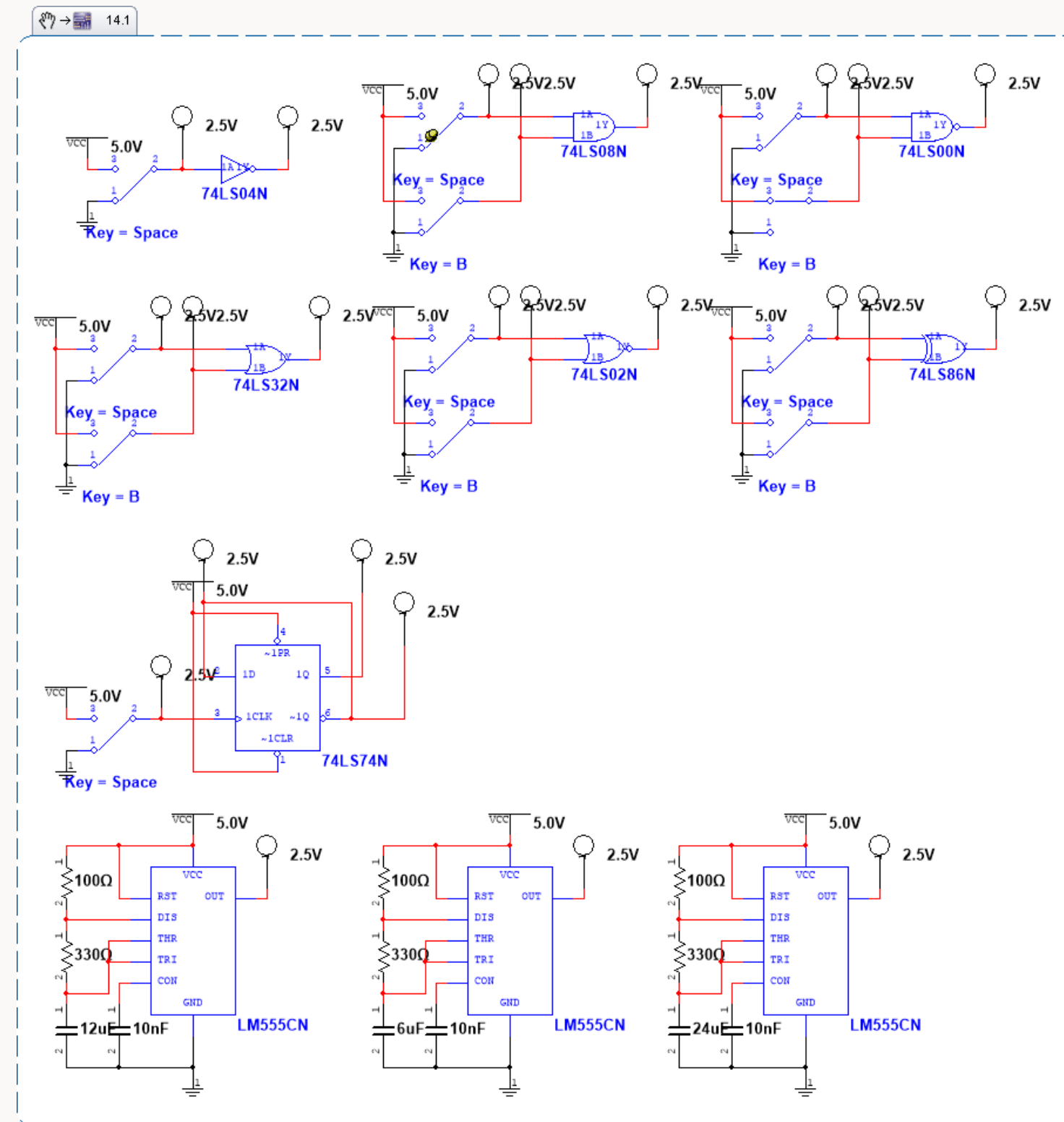
ALL EIGHT CIRCUITS, ONE SHEET

Six gate circuits, the flip-flop test circuit, and three 555 oscillators were built on a single sheet.

Every gate X-switch and the flip-flop clock switch share Key = Space; every Y-switch shares Key = B . Pressing a key toggles all switches bound to it at once.

Circuit 3's Y-switch started in the opposite position from the others and stays out of phase with them. This is cosmetic, but it is why the full-sheet probe states do not read as a single consistent pattern.

Probes read 2.5 V thresholds against a 5.00 V supply throughout.



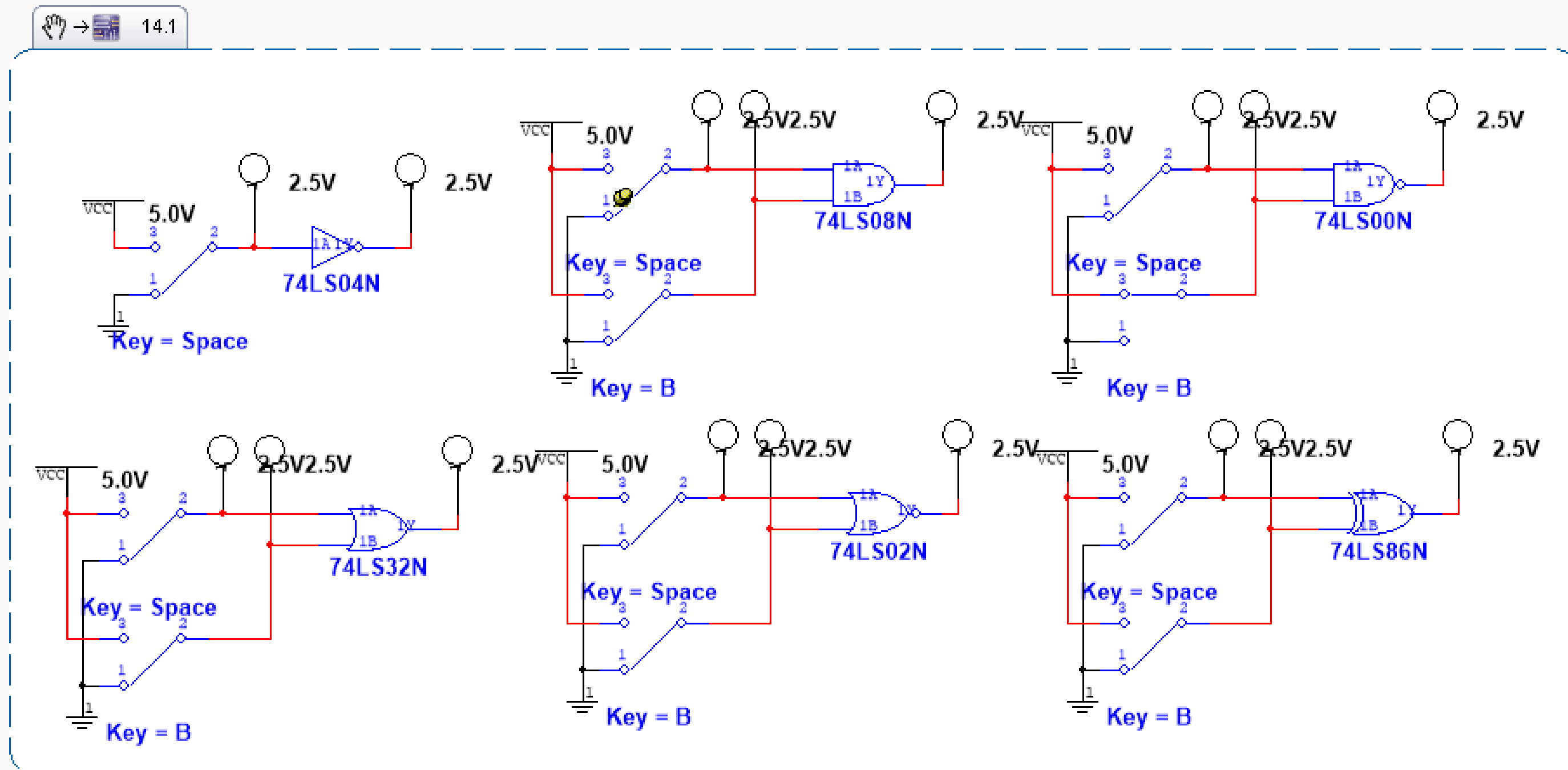
PART 1 · PROBLEMS 1–6

Combinational Logic

Six 74LS gates, all four input combinations toggled, output state recorded.

Gate Circuits as Built

PROBLEMS 1–6



Each gate is driven by SPDT switches between GND (logic 0) and +5 V (logic 1), with digital probes on both inputs and the output. All four input combinations were toggled and the output state recorded.

Six Gates, Six Truth Tables

ALL VERIFIED AGAINST SIMULATION

INVERTER74LS04N

X	Z
0	1
1	0

Output is high when the input is low, and low when the input is high.

OR74LS32N

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

Output is high when at least one input is high.

AND74LS08N

X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

Output is high only when all inputs are high.

NOR74LS02N

X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0

OR with the output inverted: high only when no input is high.

NAND74LS00N

X	Y	Z
0	0	1
0	1	1
1	0	1
1	1	0

AND with the output inverted: high unless all inputs are high.

XOR74LS86N

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

Output is high when exactly one input is high.

What the Gate Names Describe

Each name states a condition on the inputs: **all** high for AND, **at least one** high for OR, **exactly one** high for XOR. The N-prefixed parts are the same functions with the output inverted.

The definitions are phrased as “all inputs” and “at least one input” rather than a fixed count, so they hold for wider parts such as the 3-input AND.

OR VS. XOR

They agree on three rows and differ only when both inputs are high.

X	Y	OR	XOR
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	0

XOR excludes the both-high case.

PART 2 · PROBLEM 7

Sequential Logic

A 74LS74N D flip-flop wired as a toggle, observed one clock edge at a time.

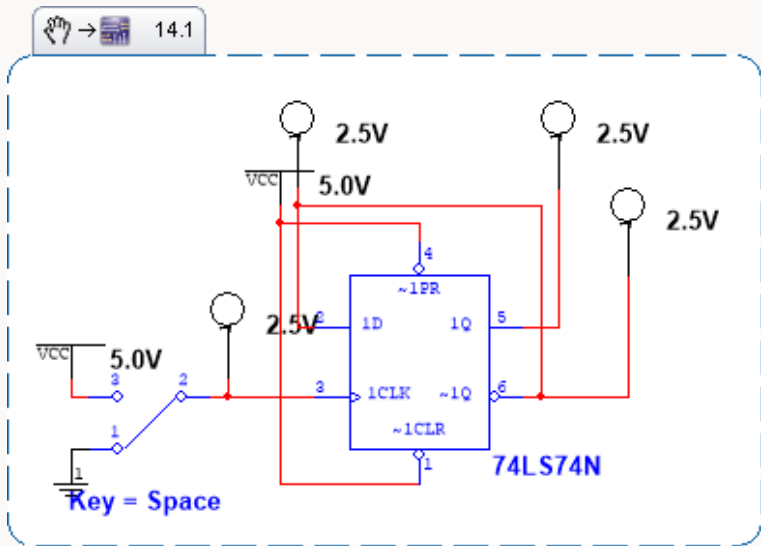
Flip-Flop Test Circuit

PIN	CONNECTION
CLK	SPDT switch, key T
D	wired to NOT_Q
PR	+5 V — held inactive
CLR	+5 V — held inactive
Probes	CLK, D, Q, NOT_Q

PR and CLR are active-low, so tying them to +5 V leaves the clock in sole control.

Starting state: the switch was toggled until CLK was low and NOT_Q was low. Every observation that follows begins there.

Because D is fed from NOT_Q, each active clock edge loads the opposite of the current output — the circuit is a toggle.



Clock Edge Behavior

b · GND → 5 V

Q turns off

High to low on the rising edge.

c · 5 V → GND

Q holds

No change on the falling edge.

d · GND → 5 V

Q turns on

Changes again on the rising edge.

Q changes at the instant CLK goes low to high — the rising edge — and not while CLK is held high. The 74LS74 is positive-edge-triggered.

Confirmed by holding CLK high for roughly three seconds: Q did not change during the interval. Because only the upward transition is active, Q changes on every other flip of the switch.

Signal Relationships

Q and NOT_Q

Always complementary

When one is high the other is low, in every state.

D and Q

Opposite here, by wiring

At each rising edge Q takes the value D held just before it. D reads opposite to Q only because D is tied to NOT_Q — in normal use D is an independent input.

Between edges

Q holds regardless of D

That holding is what makes the part a memory element — one stored bit.

A flip-flop stores one bit: it samples D on the rising clock edge, then holds that value until the next one.

PART 3 · PROBLEM 8

Clock Signals

Three LM555CN astable oscillators, identical except for the timing capacitor.

Astable Circuit and Equations

COMMON TO ALL THREE	VALUE
R_A	100 Ω
R_B	330 Ω
C1 bypass	10 nF
V_CC	5.00 V

A

12 μF

B

6 μF

C

24 μF

GOVERNING EQUATIONS, ASTABLE MODE

t_high = 0.693 (R_A + R_B) C_2

t_low = 0.693 R_B C_2

T = 0.693 (R_A + 2 R_B) C_2

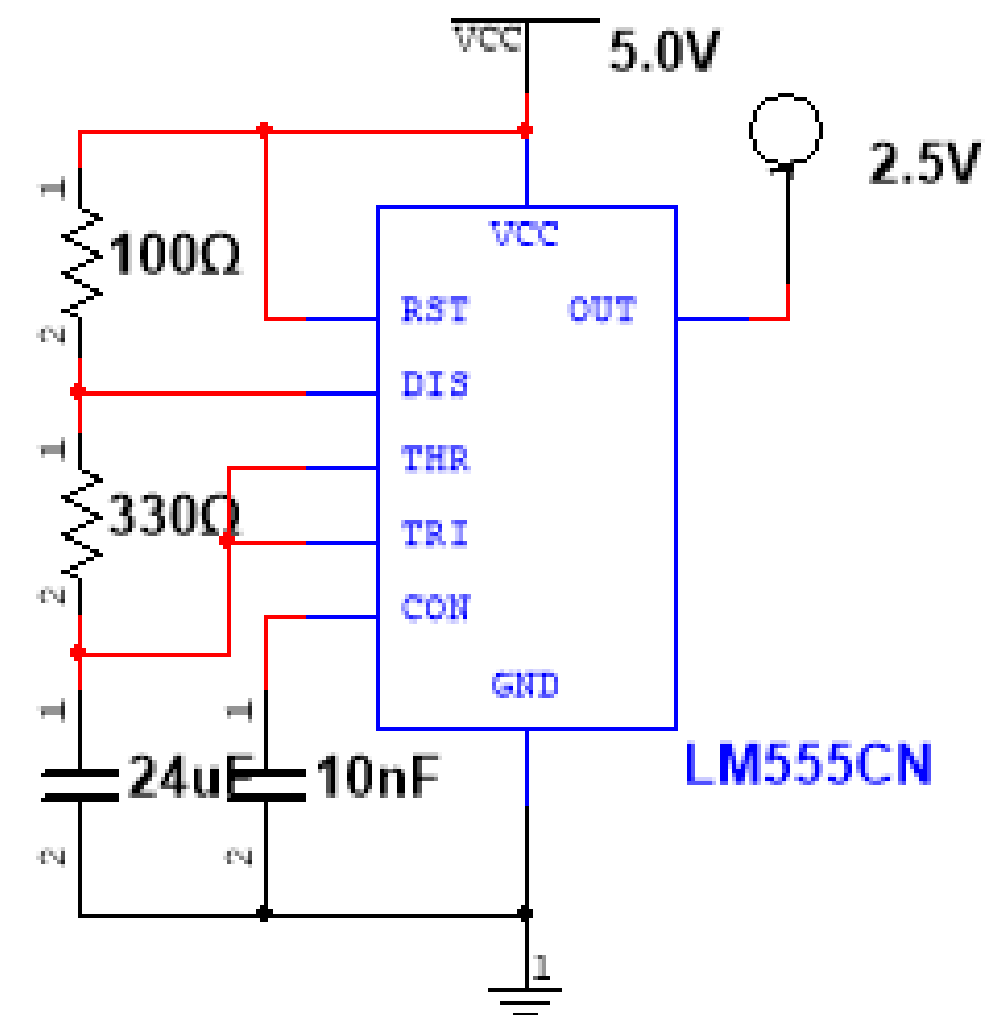
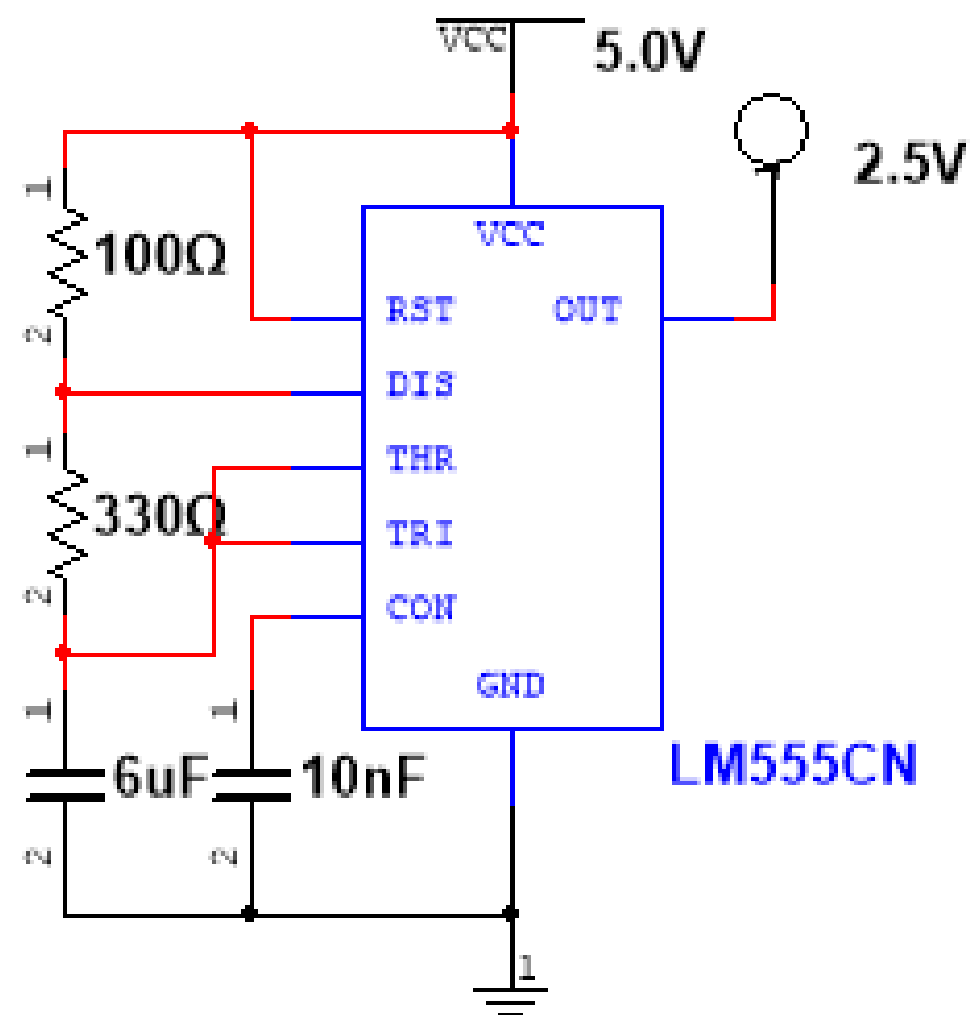
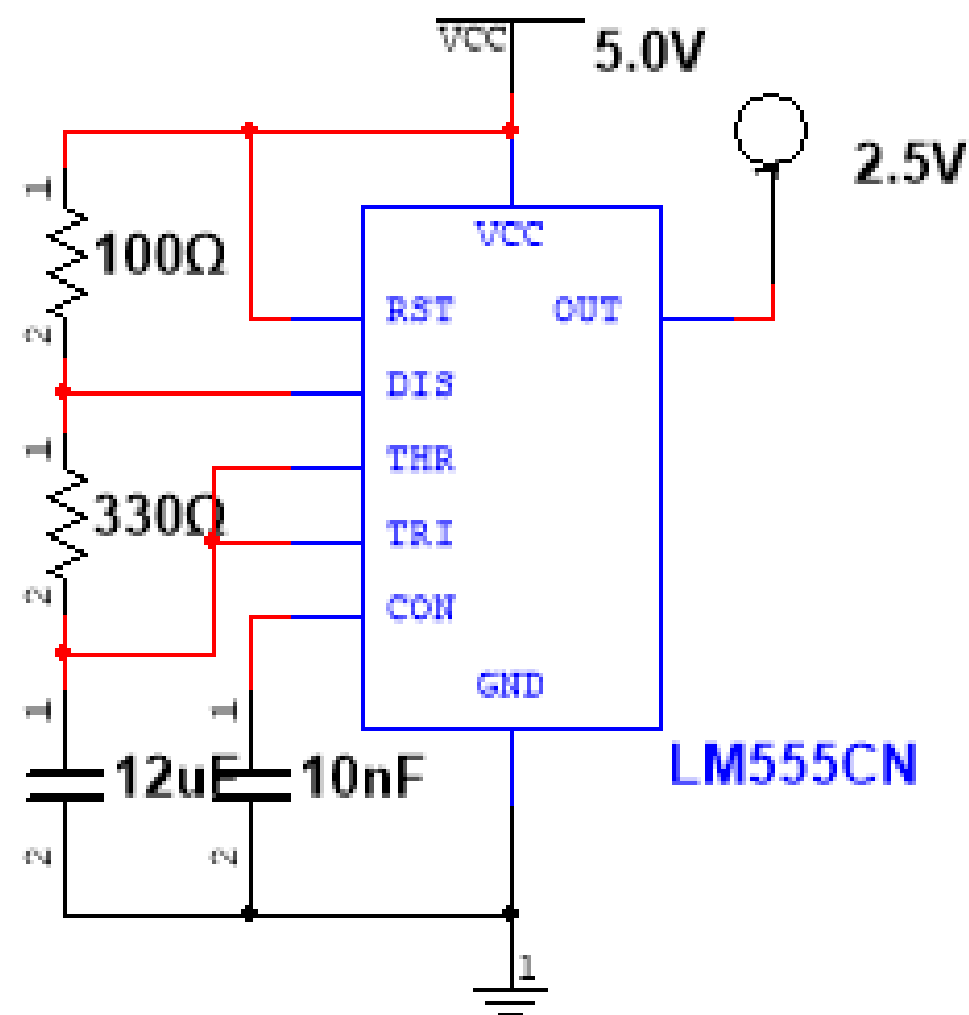
f = 1.44 / ((R_A + 2 R_B) C_2)

C_2 multiplies both intervals, so it sets speed but not duty cycle.

Oscillator Circuits as Built

PROBLEM 8 · A, B, C

14.1



All three are identical except for C_2 — 12 μF , 6 μF and 24 μF , left to right. $R_A = 100\ \Omega$, $R_B = 330\ \Omega$ and the 10 nF control-voltage bypass are common to all three. Output was measured at the OUT pin.

Measured Results

CKT	C ₂	CYCLES	T _{HIGH}	T _{LOW}	PERIOD T	FREQUENCY	DUTY
A	12 μF	13	3.577 ms	2.767 ms	6.344 ms	157.6 Hz	56.4 %
B	6 μF	21	1.789 ms	1.384 ms	3.172 ms	315.2 Hz	56.4 %
C	24 μF	5	7.154 ms	5.534 ms	12.687 ms	78.8 Hz	56.4 %

Period taken as the average interval between successive rising edges over all complete cycles, discarding the first to exclude the power-on transient. Cycle-to-cycle jitter stayed below 0.5 μs in every capture.

Measured vs. Calculated

CKT	T CALC	T MEAS	ERROR	F CALC	F MEAS
A	6.320 ms	6.344 ms	+0.38 %	158.2 Hz	157.6 Hz
B	3.160 ms	3.172 ms	+0.39 %	316.5 Hz	315.2 Hz
C	12.640 ms	12.687 ms	+0.37 %	79.1 Hz	78.8 Hz

< 0.4 %

Agreement with calculation, all three circuits

The bias is positive in all three cases and consistent in size. The ideal equations neglect the saturation voltage of the internal discharge transistor and the finite comparator switching time, both of which lengthen the charge interval slightly.

Effect of Changing C₂

$12\ \mu\text{F} \rightarrow 6\ \mu\text{F}$

Twice as fast

$T: 6.344\ \text{ms} \rightarrow 3.172\ \text{ms}\ (\times 0.5001)$

$f: 157.6\ \text{Hz} \rightarrow 315.2\ \text{Hz}\ (\times 2.000)$

$12\ \mu\text{F} \rightarrow 24\ \mu\text{F}$

Half as fast

$T: 6.344\ \text{ms} \rightarrow 12.687\ \text{ms}\ (\times 2.000)$

$f: 157.6\ \text{Hz} \rightarrow 78.8\ \text{Hz}\ (\times 0.5000)$

56.4 %

Duty cycle is unchanged in all three circuits. C₂ multiplies t_{high} and t_{low} equally, so their ratio is preserved; only R_A and R_B set the duty cycle.

RECORDED, NOT CORRECTED SILENTLY

Discrepancies Noted

01

R_A = 100 Ω is below practical minimum

The discharge transistor sinks about 50 mA through R_A on each cycle — within the 200 mA rating, so the simulation runs, but at least 1 k Ω is normal design practice.

02

The blink cannot be seen at these values

78.8–315.2 Hz is well above the ~60 Hz flicker-fusion threshold, so all three probes appear steadily lit. The worksheet most likely intended 100 k Ω and 330 k Ω , giving 6.32 s, 3.16 s and 12.64 s.

03

Reversed scope probe on two captures

The 12 μ F and 24 μ F traces swing –5 V to 0 V. Display is inverted; edge timing, period and duty cycle are unaffected. The 6 μ F capture reads 0 V to +5 V.

The circuits were built as printed and measured on the oscilloscope. Both comparison answers depend only on ratios, so they hold either way.

Status: Content Complete

DONE

- All eight problems answered and verified
- Six truth tables confirmed against simulation
- Problem 7 a–g complete; b, c, d cross-checked on a screen recording
- Problem 8 measured from raw scope data, within 0.4 %
- Three discrepancies documented

OUTSTANDING

- Name and date are still placeholders in the header
- Multisim captures exist but are not yet cropped per-circuit in the report itself
- No oscilloscope trace images — numbers came from the .scp files
- LaTeX timing equations need conversion if the target does not render math

Right, Wrong, Learned

DONE RIGHT

- Every problem answered and checked against simulation
- Circuits built with the printed values instead of quietly corrected
- Period averaged over full cycles with the power-on transient discarded
- Edge-triggering claim verified on a screen recording, not assumed
- Units and engineering notation kept consistent throughout

DONE WRONG

- Scope probe reversed on two of the three captures
- All six gate X-switches bound to the same key, so inputs toggle together
- Circuit 3's Y-switch left out of phase with the others
- Scope traces never exported as images — only the numeric .scp data was kept
- Name and date left blank in the report header

LEARNED

- Combinational output depends only on present inputs; sequential output depends on clock history
- Holding a value between clock edges is what makes a flip-flop a memory element
- In a 555 astable, C_2 scales the period while the resistor ratio fixes the duty cycle
- Ideal equations run about 0.4 % fast because they ignore real switching losses
- Above flicker fusion a blinking output is indistinguishable from a steady one — measure, don't look
- Raw capture files need transients, polarity and dead channels handled before any measurement

CONCLUSION

Seven Components, Three Classes

Six 74LS gates

Output determined entirely by the present state of the inputs, with no dependence on history.

74LS74N

Output depends on when the clock transitions, and is retained between edges — a one-bit memory.

LM555CN

Supplies the clock such devices require, with period set by two resistors and a capacitor.

Combinational logic, sequential logic, and a clock source are the building blocks for the Random Number Generator designed later in this unit.